METABOLISM

Module map

- Module 06: Metabolism
- Connected lab: lab-07_metabolism.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 06
Metabolism

1. Energy and thermodynamics
2. ATP as cellular coupling
3. Enzymes and activation energy
4. Regulation by conditions and inhibitors
5. Photosynthesis and respiration as linked pathways

Lab evidence
lab-07_metabolism.md

METABOLISM

Learning objectives

- Apply thermodynamic ideas to living cells.
- Explain ATP as an energy-coupling molecule.
- Describe how enzymes change reaction rates.
- Predict how environmental conditions affect enzyme activity.
- Connect photosynthesis and cellular respiration in matter and energy flow.

OBJECTIVE 1
Apply thermodynamic ideas to living cells.

OBJECTIVE 2
Explain ATP as an energy-coupling molecule.

OBJECTIVE 3
Describe how enzymes change reaction rates.

OBJECTIVE 4
Predict how environmental conditions affect enzyme activity.

OBJECTIVE 5
Connect photosynthesis and cellular respiration in matter and energy flow.

METABOLISM

Topic sequence

- Energy and thermodynamics: Cells transform energy while obeying thermodynamic constraints.
- ATP as cellular coupling: ATP couples energy-releasing reactions to energy-requiring cellular work.
- Enzymes and activation energy: Enzymes lower activation energy without being consumed.
- Regulation by conditions and inhibitors: Temperature, pH, substrate concentration, and inhibitors alter enzyme function.
- Photosynthesis and respiration as linked pathways: Photosynthesis builds sugar while respiration breaks fuel down to harvest usable energy.

01

Energy and thermodynamics

Cells transform energy while obeying thermodynamic constraints.

02

ATP as cellular coupling

ATP couples energy-releasing reactions to energy-requiring cellular work.

03

Enzymes and activation energy

Enzymes lower activation energy without being consumed.

04

Regulation by conditions and inhibitors

Temperature, pH, substrate concentration, and inhibitors alter enzyme function.

05

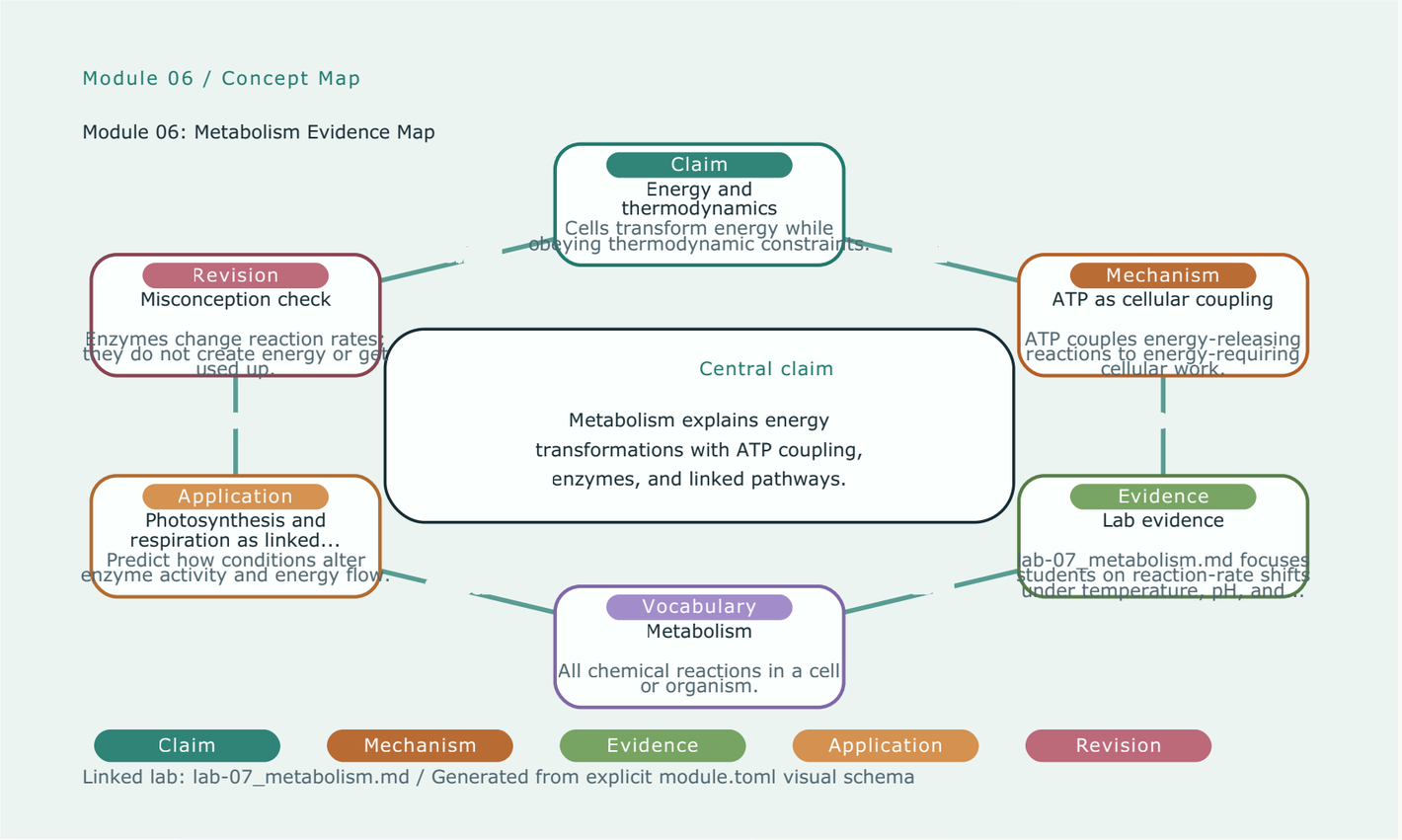
Photosynthesis and respiration as linked pathways

Photosynthesis builds sugar while respiration breaks fuel down to harvest usable energy.

METABOLISM

Module 06: Metabolism Evidence Map

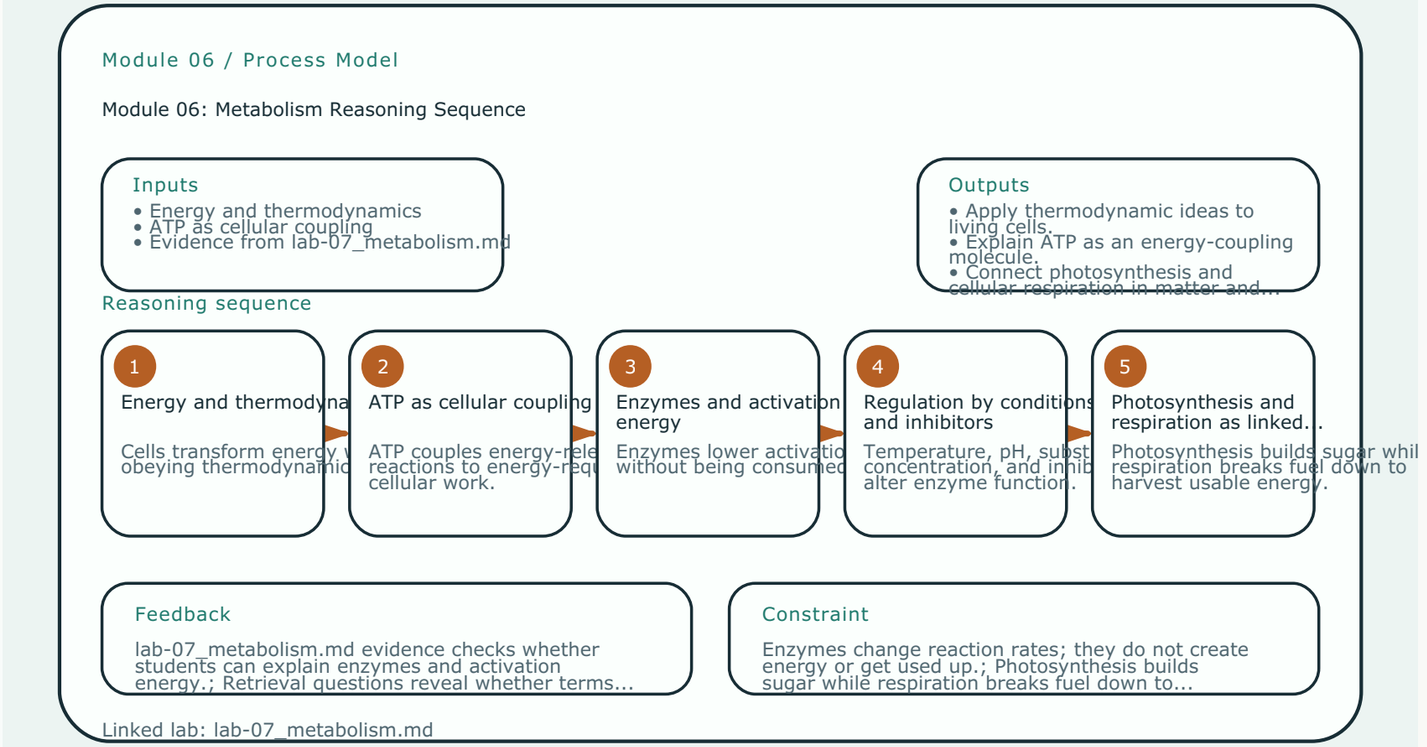
- Module focus: Metabolism
- Central claim: Metabolism explains energy transformations with ATP coupling, enzymes, and linked pathways.
- Key nodes to connect: Energy and thermodynamics, ATP as cellular coupling, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-07_metabolism.md supplies an evidence surface for the map.



METABOLISM

Module 06: Metabolism Reasoning Sequence

- Module focus: Metabolism
- Trace the model: Energy and thermodynamics -> ATP as cellular coupling -> Enzymes and activation energy -> Regulation by conditions and inhibitors
- Inputs become outputs: Energy and thermodynamics, ATP as cellular coupling -> Apply thermodynamic ideas to living cells., Explain ATP as an energy-coupling molecule.
- Feedback check: lab-07_metabolism.md evidence checks whether students can explain enzymes and activation energy.
- Lab connection: lab-07_metabolism.md gives students a process to observe or test.



METABOLISM

Terms as evidence handles

- Metabolism: All chemical reactions in a cell or organism.
- ATP: Cellular energy-coupling molecule.
- Activation energy: Energy required to start a reaction.
- Enzyme: Protein or RNA catalyst that speeds a reaction.
- Substrate: Reactant that binds an enzyme active site.
- Denaturation: Loss of shape that disrupts function.

METABOLISM
All chemical reactions in a cell or organism.

ATP
Cellular energy-coupling molecule.

ACTIVATION ENERGY
Energy required to start a reaction.

ENZYME
Protein or RNA catalyst that speeds a reaction.

SUBSTRATE
Reactant that binds an enzyme active site.

DENATURATION
Loss of shape that disrupts function.

METABOLISM

Lab connection

- Lab file: lab-07_metabolism.md
- Lab evidence should test or illustrate: Enzymes and activation energy
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Energy and thermodynamics

EVIDENCE
lab-07_metabolism.md

CLAIM
Apply thermodynamic ideas to living cells.

METABOLISM

Contrast check

- Do not stop at: Cells transform energy while obeying thermodynamic constraints.
- Stronger explanation: Photosynthesis builds sugar while respiration breaks fuel down to harvest usable energy.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

Cells transform energy while obeying thermodynamic constraints.

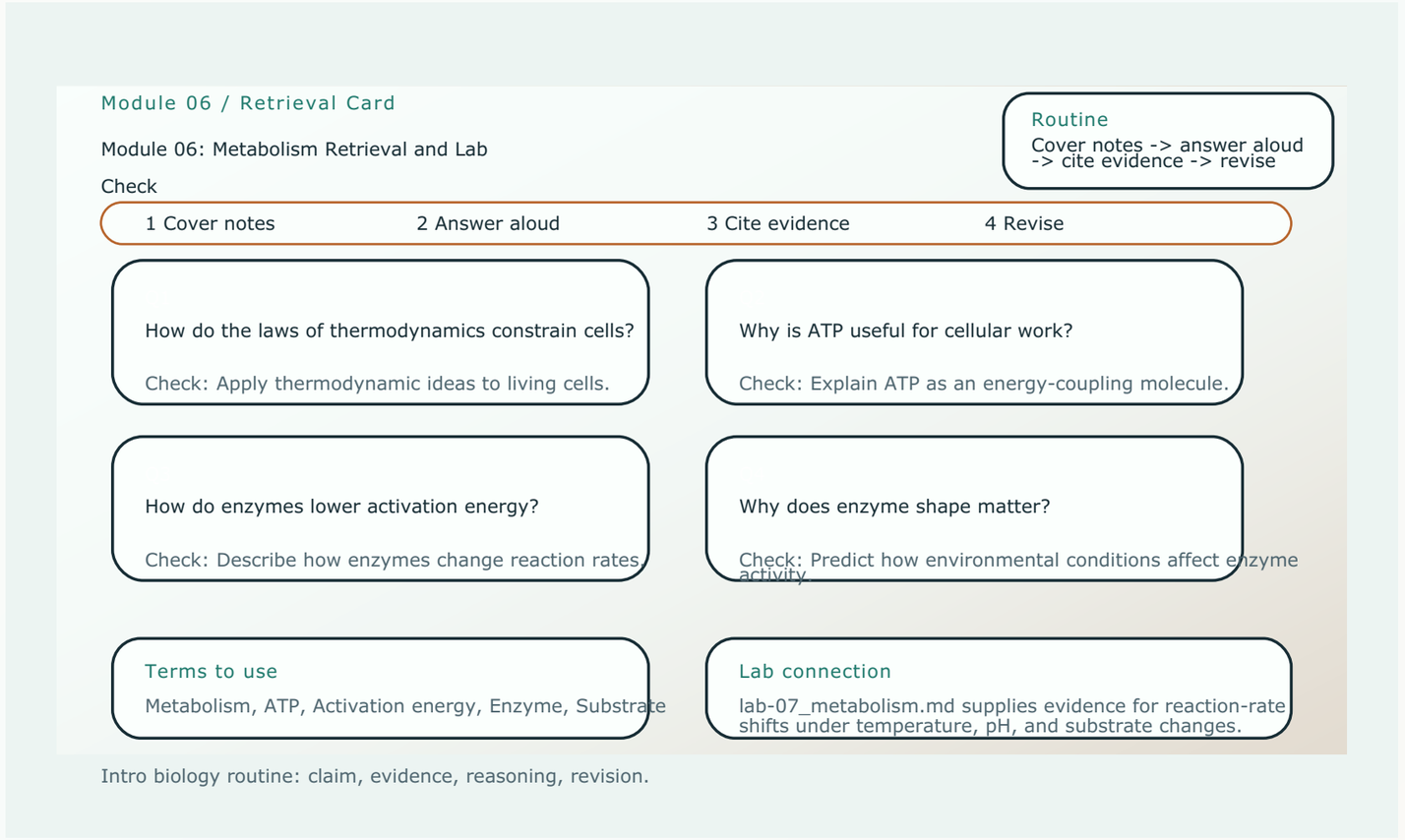
DEEPER

Photosynthesis builds sugar while respiration breaks fuel down to harvest usable energy.

METABOLISM

Module 06: Metabolism Retrieval and Lab Check

- Module focus: Metabolism
- Retrieval prompt: How do the laws of thermodynamics constrain cells?
- Answer check: Apply thermodynamic ideas to living cells.
- Required terms: Metabolism, ATP, Activation energy
- Lab connection: lab-07_metabolism.md; lab-07_metabolism.md supplies evidence for reaction-rate shifts under temperature, pH, and substrate changes.



METABOLISM

Practice quiz bridge

- 1. What is the main role of ATP in a cell?
- 2. How does an enzyme speed up a chemical reaction?
- 3. Raising temperature far above normal can stop an enzyme from working because:
- 4. How are photosynthesis and cellular respiration related?

QUESTION 1

Answer first

What is the main role of ATP in a cell?

QUESTION 2

Answer first

How does an enzyme speed up a chemical reaction?

QUESTION 3

Answer first

Raising temperature far above normal can stop an enzyme from working because:

QUESTION 4

Answer first

How are photosynthesis and cellular respiration related?

METABOLISM

Synthesis and exit ticket

- Exit claim: explain energy and thermodynamics using evidence from lab-07_metabolism.md.
- Must include: Metabolism, ATP, Activation energy.
- Revision prompt: what evidence would change your explanation?

CLAIM
Energy and thermodynamics

EVIDENCE
lab-07_metabolism.md

REVISION
What would change your mind?